



Society of Petroleum Engineers

SPE-229586-MS

Realizing the Need for Agentic AI for Subsurface Data and Workflows

Pranav Kumar, Bedoor Saeed Al Mazrouei, Muhammad Faisal Fayyaz, Salahuddin Abdullah Jan, and Jawhara Al Maysari, ADNOC ONSHORE, Abu Dhabi, United Arab Emirates

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229586-MS](https://doi.org/10.2118/229586-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

Subsurface data consists of various types, and it is used in various business workflows. The efforts spent on performing data checks and running the workflows are extensive. Advancements in machine learning and natural language processing have been beneficial in extracting information from unstructured data. Especially, the AI Agents now play very useful roles and are capable of performing repetitive tasks. This abstract presents some of the Agentic AI supported concepts, where data is extracted, quality checked and used in Subsurface workflows.

One example of Agentic AI workflow, where the information is extracted and data is made searchable from unstructured Cased Hole logs reports. These well logs support decisions related to Well Production. Following Steps are performed to achieve the use case objective.

- Data Extraction: Automatically extracting text and data from scanned documents and other documents (pdf, word, excel)
- Digital Data Availability: Comparing data from documents with digital data formats like LAS and DLIS.
- Natural Language Processing: Using NLP to understand query intent from available tools.
- Large Language Models: Using available LLM models and developing & training new models using extracted information.
- Retrieval Augmented Generation: Enhancing model accuracy by retrieving reliable data sources like SharePoint or custom file locations.
- Web User Interface: Allowing users to interact with AI agents for information retrieval.

Having an Agentic AI chatbot developed using Large Language model, extracts information from the knowledge source and then provides information formatted using LLM to the business users. The chatbot helps support data management engineers in quality checking the data. This model is trained based on techniques like Summarization, prompt and classification.

As a part of the presented work, various file types for the cased hole logs were collected with focus on categories like Production logging tool, Reservoir Saturation tool and Injection Logging tool. After

Running the Natural language Processing techniques, different sections of the unstructured documents were extracted. A large language model to be developed based on the extracted information. To keep the LLM updated with the new data, Retrieval Augmented Generation is used. One of the examples of Chatbot for unstructured data prompts is mentioned here:

The chatbot answers the following questions:

Summarize the Well ****NAME***

Overview of Well ****NAME***

Which Wells have RST data

Which Wells have PLT data

Which Wells have ILT data

Which Section contains cased hole data

Show me the tool names

Show me the plots

What are the available Cased Hole tools available in Well ****NAME***

Show me the data available in Reservoir Name

Show me the Well Sketch of Well ****NAME***

Show the Well Schematic of Well ****NAME***

Depending on the number of wells and required Petrophysical processing for any Subsurface analysis, it takes months to completed the workflow. Business users must go through all the unstructured data and reports. This is a very time-consuming process. Having a chatbot agent to find relevant information and a quality checked database support business workflows. In future, the large language model can be fine-tuned to incorporate more data categories. Utilization of Generative AI techniques supports the business and keeps the information up to date.

Introduction

The subsurface domain encompasses a complex ecosystem of various data types, workflows, databases, web portals, and dashboards. The information within this sector exists in a range of formats, ranging from unstructured (such as scanned reports, handwritten notes, and free-text documents), to semi-structured (like spreadsheets and logs), and fully structured formats (including relational databases). Numerous Petro-Technical applications are employed to execute specialized subsurface workflows, each generating and requiring access to their own data sets.

Agentic AI tools have rapidly emerged as valuable allies in navigating and extracting relevant insights from these diverse and often fragmented data sources. By leveraging advanced prompt-based queries and workflow automation capabilities, these tools can significantly streamline the process of locating, analyzing, and synthesizing critical information.

For example, consider the process of Petrophysical data analysis. If a Petrophysicists wants to review cased hole logs data, they would have to go through various systems like SharePoint, file systems, corporate databases and download the data along with reports. The next step involves opening each document and loading the associated data in the Petro-Technical software for review. In some cases, there is a metadata management portal that helps identifying the data and reports, but this is only beneficial if all the data sources are integrated in the portal.

The evaluation of an existing process was undertaken as a challenge by data management. The detailed assessment of available subsurface data sources and documenting the data flow from the raw to final stage was conducted.

Method	Participants	Key Pain Points Identified	
Interviews and workshops	Domain experts and business users	Manual searching, lack of interoperability, time lost in cross-referencing across disconnected systems	

Solution	Capabilities	Examples of Queries	How It Works
Agentic AI	Intelligent agent digital assistants, understand domain-specific queries, perform tasks autonomously	"Which wells have cased hole data available in a particular reservoir?", "Show me the well schematic for Well X."	Indexing and interpreting content from various file types, consolidating results, that can be presented in a specific format.

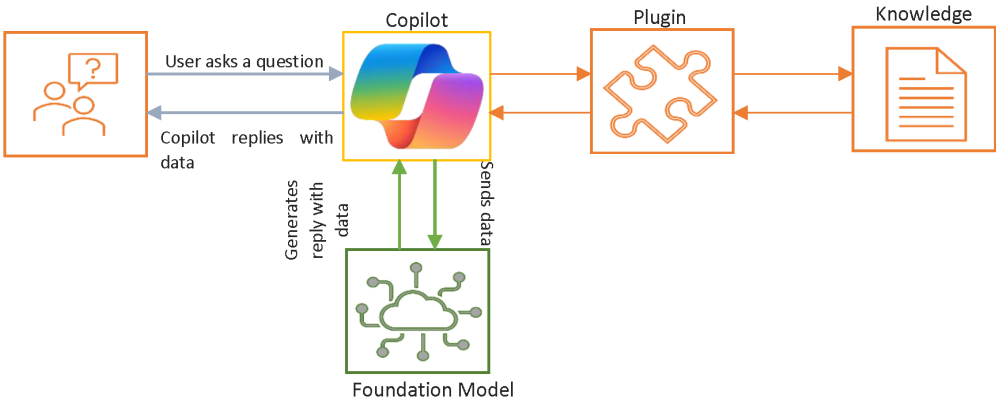


Figure 1—The copilot studio agent architecture shows the prompt-response agent for answering queries on the basis on Knowledge source provide to the agent. The LLM used in the copilot studio is GPT-4.

To start the work, several small business cases were created, and Proof of Concept were executed. There were the following challenges encountered at the start:

- **Buy-In from The Stakeholders:** There are various stakeholders from respective Asset teams, management, Information Technology and data management. The Objective of proof of concept was presented to the stakeholders with a user story, showcasing the layout for the AI Agent and the configuration window from the Copilot Studio interface. A comparison of manual and AI workflow was presented.
- **Required Infrastructure:** The Proof of concepts require software, hardware and custom requirements, where required Open -Source libraries python libraries can be downloaded and configured. To overcome this challenge the existing Information Technology department was involved throughout the process. For the Copilot Studio program, private cloud environment was configured.
- **Data variation:** The variety of data types is the biggest challenge and an opportunity to implement GEN AI via AI agents to extract value. The agents were deployed over unstructured datasets to retrieve information about subsurface data types with simple natural language queries. These agents not only returned precise answers, but could also generate visualizations, such as well plots or schematics, directly from the underlying data.
- **Similar Agents:** The size of the data presents the major challenge as agents are being created, and the users are kept on uploading the same documents. To overcome this, it was suggested to the user to have a common agent for a team and only one person upload the data and refresh the agent.
- **Vector database and Dataverse:** One of the challenges that is predicted in the future is related to the growth in the size of the unstructured data. This must be overcome by implementing vector databases and Dataverse. These are currently evaluated as a part of the Proof of concepts for Agentic AI.

The results of the Proof of Concepts, with the creation of specific AI Agents, attributed the following objectives:

Benefit	Description
Efficient information extraction:	The amount of time taken for extracting information from multiple documents was reduced from days to hours.
Enhanced data quality	By setting up data quality rules and asking proper prompts, the data quality has improved. Over the period, the data quality has become easier to manage
Stakeholder collaborations	Having a Single type of AI agent via a dedicated platform allows geoscientists, engineers, and business analysts to use single source of data
Be Ready for Future	The conducted Proof of concept and Gen AI works as a foundation for any integrations required for the AI implementation.

Subsurface data management workflows and its integration with Agentic AI make it easier for organization to deal with a variety of data sets. By standardizing the prompts and using the standard agents with similar backend environment, producing information from disparate sources, and providing in-built interfaces for business users, these tools are controlled to become vital in optimizing business workflows and decision-making in the forever evolving landscape of Subsurface data management.

Statement of Theory and Definitions

A **copilot studio** is an AI-powered assistant that provides support for tasks, offers insights, and boosts productivity.

An **agent** is a specialized AI tool designed to handle specific processes or solve business challenges. These agents can automate tasks, analyze data, make decisions, and adapt to new challenges. They operate independently or in conjunction with a copilot studio, which acts as the interface for interacting with these agents.

Large language models (LLMs) are the core technology powering chatbot agent copilot studios, enabling them to understand natural language, generate human-like text, and perform various tasks. These LLMs can be used to power the core functionality of a chatbot agent, including responding to user input, generating code suggestions, and even automating tasks. Different platforms and tools offer various ways to leverage LLMs for these purposes, from pre-built agents to custom agent development. The LLM'ss to provide various AI-powered features across its applications, including summarizing documents, generating emails, and scheduling meetings.

Langchain is a framework for building application based on large language models (or LLMs). It allows us to use multiple LLM in our applications. It supports integrating the LLM with local data sources using Chains generate responses. Langchain components allow LLM's to access new datasets without retraining. A chain is a series of automated actions from the user's query to the model's output. It is used to connect different data sources, generate unique content and answer queries.

Methodology

Subsurface Data Sources

The categorization of subsurface data is tied to its sources, with unstructured data representing most of the information within the subsurface domain. Unstructured data includes a wide array of document formats such as PDFs, PowerPoint presentations, and Word files, that often lack consistent formatting and may be scattered across numerous repositories. To address this challenge, AI Agents leverage prompt-and-response mechanisms, powered by sophisticated large language models, to intelligently extract, interpret,

and deliver relevant information in response to user queries. These agents essentially transform raw, unstructured knowledge bases into accessible, actionable insights by sifting through diverse document types with remarkable efficiency.

The core business case for such an AI Agent centers on the extraction of critical information from collections documents and reports. In the upstream oil and gas sector, these reports are vital for understanding well integrity, reservoir characteristics, and production dynamics.

Cased Hole Insight Agent. The first business case mentioned here is the creation of an Agent for querying unstructured case hole logs reports and data. The reports typically contain data from various specialized logging tools, including injection logging tools, production logging tools, and reservoir saturation tools. Commonly acquired logs such as Sigma, Gamma Ray, and Pulse Neutron logs provide crucial subsurface measurements, while additional log curves further augment the dataset required for comprehensive analysis.

For petrophysicists and geoscientists, processing and interpreting data across multiple wells is a complex, time-consuming endeavor. Traditionally, each report, along with its associated data files (such as LAS and DLIS), must be manually reviewed, quality-checked, and cross-referenced with other datasets. If discrepancies or missing information are discovered, the process often involves reaching out to service providers for clarification, introducing further delays. This challenge is particularly pronounced when dealing with legacy wells, whose historic records may be incomplete, inconsistently formatted, or difficult to locate within vast archives.

AI Agents mitigate these challenges by automating the process of document review and data extraction. By recognizing standard report structures and parsing key information, they reduce the need for manual intervention. For example, most cased hole log reports adhere to a consistent layout:

Section	Description
Header Section	Contains metadata about the service provider, operating company, well, field, rig, country, location, logging date, run number, datum, logging interval, and responsible personnel.
Comments	Details of the logging objectives, remarks about the tool's performance, and downhole well status observations.
Tool Image	Supplies diagrams and descriptions of the instruments used during logging, aiding in identification and interpretation.
Log Plots	Displays graphical representations of the acquired log curves along with processed parameters and post-acquisition analyses.

By extracting and organizing these sections, AI Agents ensure users can swiftly access the precise information they need, whether for regulatory compliance, reservoir analysis, or operational planning. The resulting workflows not only enhance productivity but also improve data quality, governance, and collaboration by centralizing validated and up-to-date knowledge in a single, easily accessible system. As AI Agents continue to evolve, their integration into subsurface data management is set to unlock advanced analytics capabilities, drive predictive modeling, and enable more informed, agile decision-making throughout the energy sector.

A Prompt and response AI Agent was created using copilot studio and the following instructions were given to the Agent: Summarize the content of the uploaded PDF files with a section-wise breakdown. Focus on extracting key information from each section, maintaining the original structure and headings where possible. If multiple files cover overlapping topics, consolidate the information under common sections. Ensure the summary is clear, concise, and suitable for quick reference.

The reports were uploaded on a SharePoint, and it was used as the Knowledge on the Agent creation. The agent was able to respond to the following prompts:

Summarize the Wells

- The list of Wells along with the Tool used, Production, Key zones, along with the depths, Surface Rate, Crossflows (if detected)

Summarize the Well **101V*

- General Information like location, Company, Tool Used, logging date, Completion and Total Depth. Objectives of the Well to detect and record pressure temperature profiles, detect fluid flow etc., Key findings including Downhole Production, Surface Rates, Water Entry, Nonproducing Interval information

Compare reports and LAS data of Well **101V*

- The comparison will list down the file names, which it is using for comparison and a Curve Comparison table will appear, showing comparison

Show Zonal distribution of Well **101V*

- The Zonal contribution table for the Well shows Columns related to “Producing Interval, Water Rate, Oil Rate, Gas rate.

Compare with Other Wells

- In the same field, a Zonal Production comparison table appears which shows all the available wells metrics.

Extract Well Schematic

- Downloadable images are extracted from the reports

Show list of Well information belonging to Reservoir

- A tabular format of all the Wells under a reservoir and their Completion, production information.

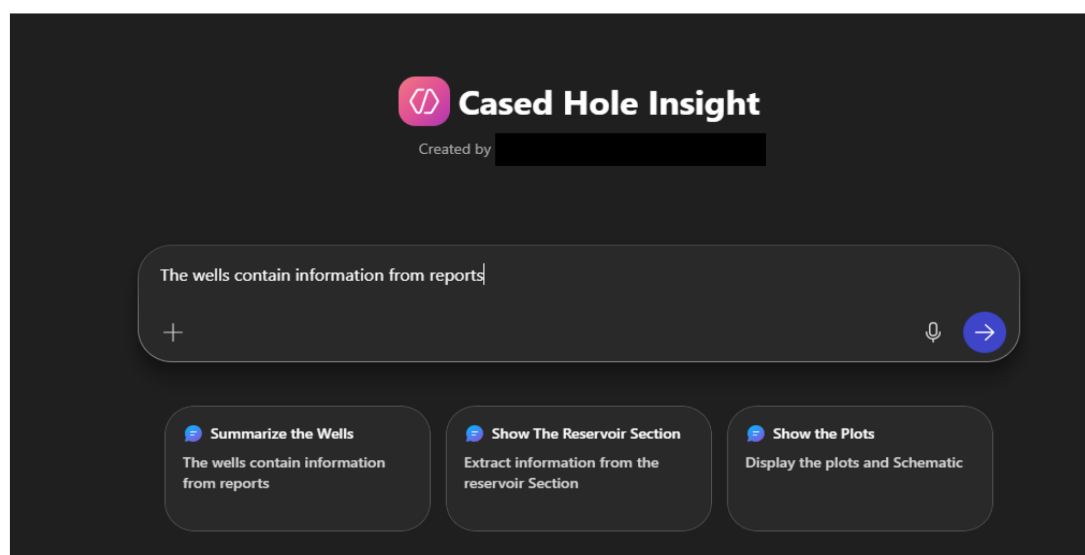


Figure 2—Cased hole Insight agent Interface is suggested prompts

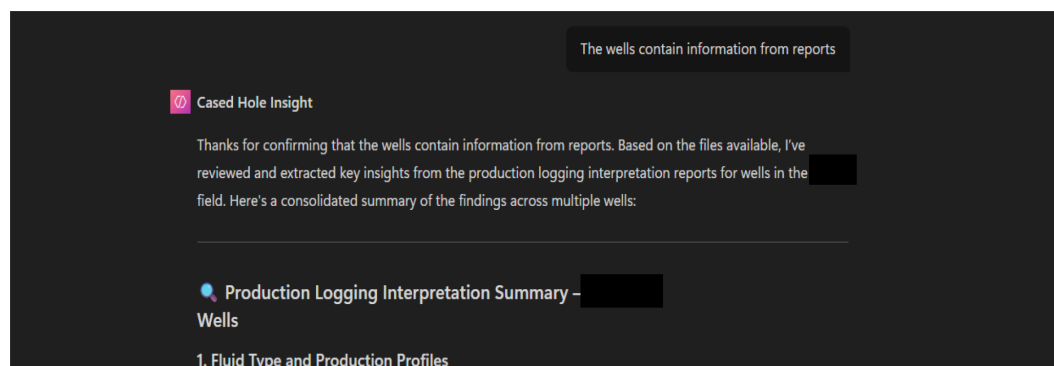


Figure 3—Cased hole Insight agent workflow shows the prompt-response agent for answering queries from unstructured cased hole logs and data

VSP Insight Agent. The second business case is utilizing Vertical Seismic Profile data and reports. VSP's contains the seismic data recorded in a borehole, providing detailed information about subsurface structures and properties. There are various file types, acquisition reports, time-depth tables, wavelet images and other kinds of data, reports and plots. A typical report of VSP contains Well information, acquisition parameters, elevation, survey information, downhole equipment information, Surface sensor configuration, Sonic calibrations processing, Log conditioning, Synthetic Seismogram Processing, Polarity Convention.

This agent focuses on leveraging Vertical Seismic Profile (VSP) data and the associated reports. VSPs are specialized seismic measurements taken from within a borehole, offering a more precise depiction of subsurface geological structures and their physical properties compared to surface seismic surveys. These datasets are invaluable in reservoir characterization, well placement, and seismic calibration.

A comprehensive VSP report typically includes:

Category	Description
Well Information	Detailed identification of the well, its location, target zone, and drilling context.
Acquisition Parameters	Specifications of the seismic survey, such as source type, receiver array, geometry, and sampling rates.
Elevation Data	Precise elevation references for both surface and downhole sensors, essential for accurate depth conversion and time correlation.
Survey Information	Overview of the survey design, objectives, and any site-specific challenges encountered during acquisition.
Downhole Equipment Information	Details on the sensors, cables, and tools deployed within the well, including their locations, sensitivities, and configurations.
Surface Sensor Configuration	Arrangement and calibration of geophones or accelerometers placed at the surface to support data integration.
Sonic Calibrations Processing	Procedures for calibrating sonic tools and aligning sonic log data with VSP measurements to improve subsurface velocity models.
Log Conditioning	Preprocessing steps applied to raw logging data to remove noise, enhance signal quality, and prepare for interpretation.
Synthetic Seismogram Processing	Generation of synthetic seismic traces from well log data, enabling comparison with field VSP and surface seismic responses.
Polarity Convention	Documentation of the polarity standards used, ensuring consistency between VSP results and broader seismic datasets.

Collectively, these components provide a robust foundation for integrating borehole seismic insights into reservoir modeling and subsurface interpretation workflows. VSP data not only improves the vertical resolution of seismic images but also enhances the calibration of surface seismic, leading to more reliable geological models and more informed exploration and production decisions.

An AI Agent was created using Copilot studio. List of VSP data and reports were uploaded to a SharePoint, which was used as a knowledge in the copilot studio.

The agent was able to respond to the following prompts:

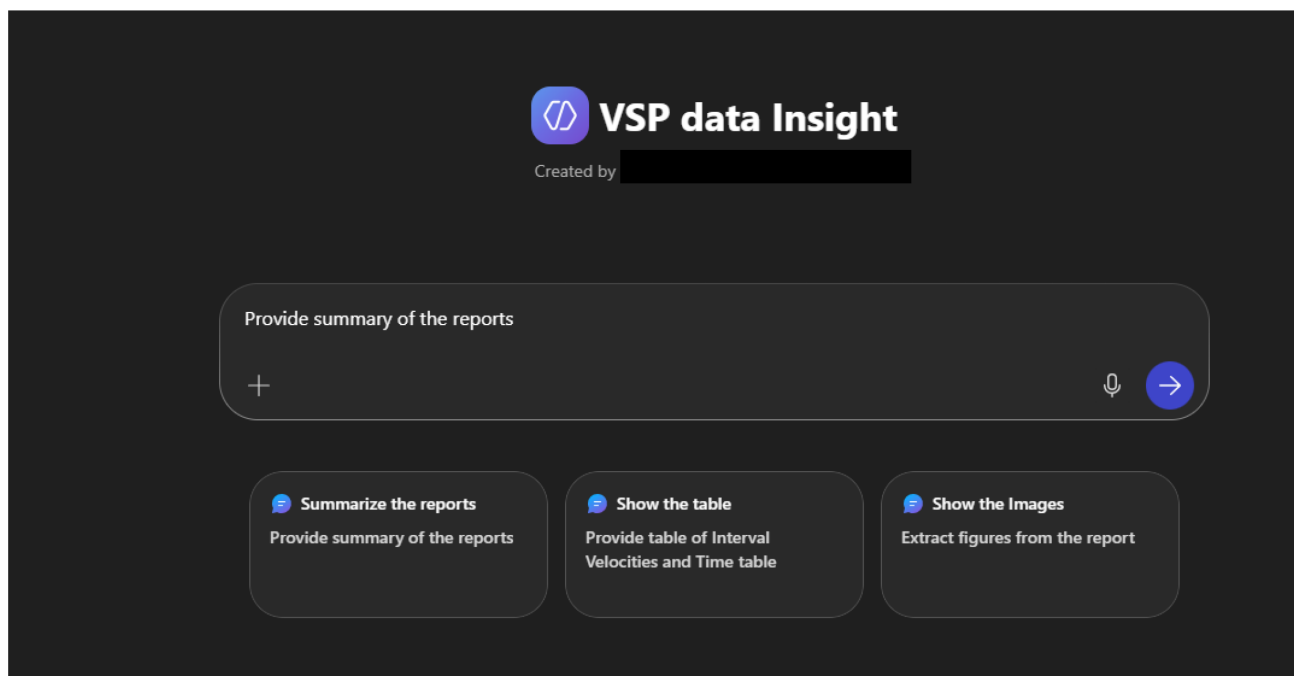
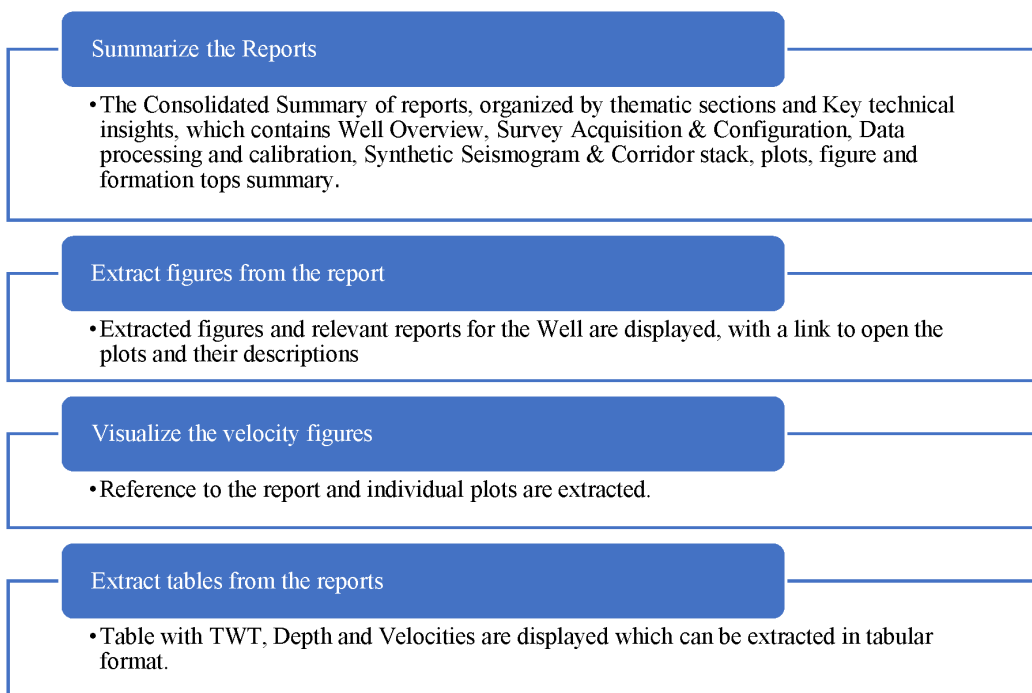


Figure 4—VSP data insight agent interface

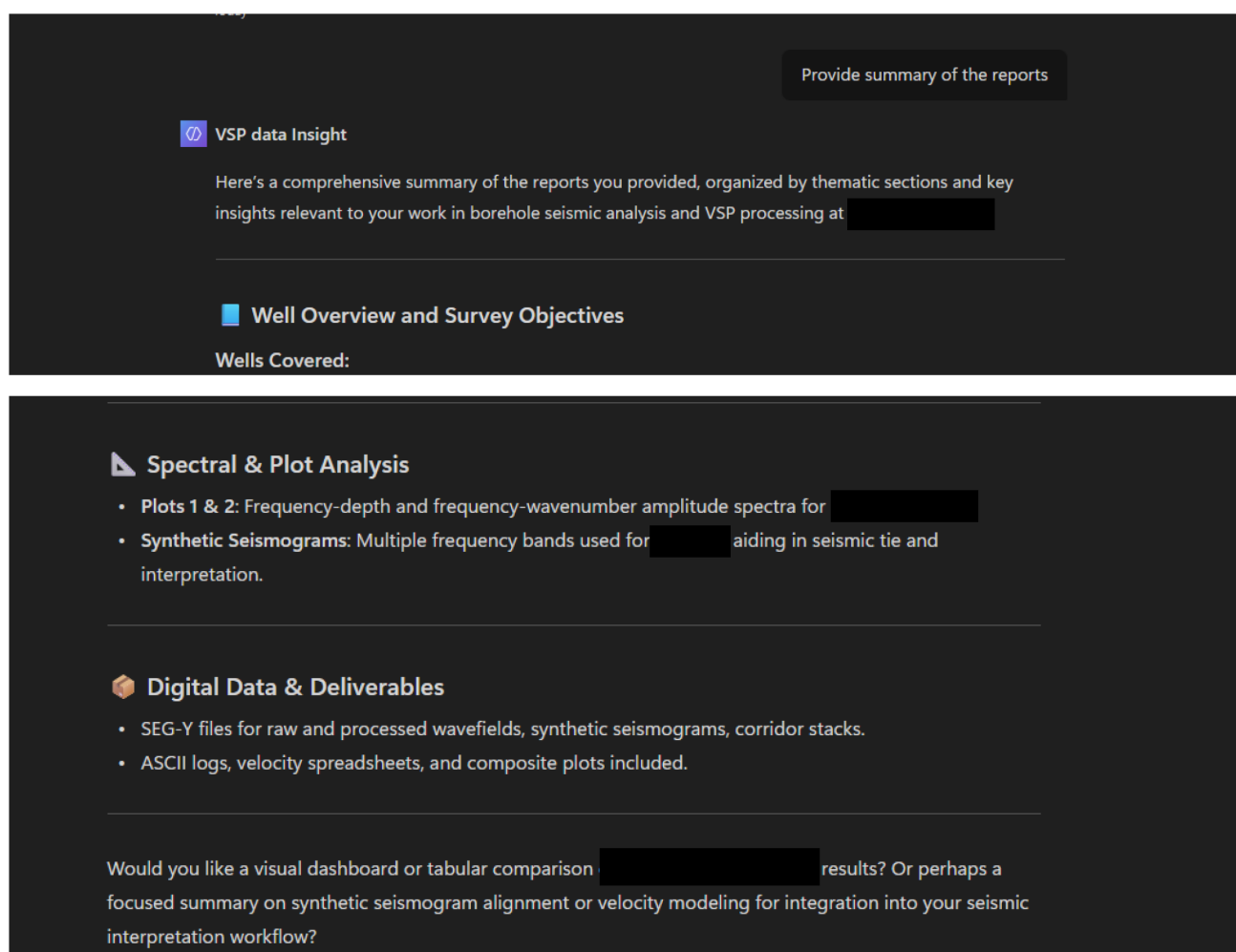


Figure 5—VSP Data Insight agent workflow shows the prompt-response agent for answering queries from unstructured reports

Sedimentologist Insight Agent. The third business case covers information insight from the Sedimentological and Petrographical analysis reports. These reports provide a comprehensive overview of core samples, focusing on their composition, texture, and depositional characteristics. Detailed core descriptions are meticulously prepared to classify the lithological facies identified by their physical, mineralogical, and sedimentary features. These facies are then interpreted to reconstruct depositional environments, such as fluvial, deltaic, or marine settings, offering insights into the geological history and reservoir quality.

In addition to qualitative descriptions, the reports integrate quantitative data derived from Petrographic analysis, which involves microscopic examination of thin sections to identify mineral content, grain size distribution, porosity, and diagenetic alterations. This information is complemented by results from conventional core analyses, such as porosity, permeability, and grain density measurements.

A key aspect of the reports is the seamless integration of core description logs with wireline log data. This correlation allows for a more robust interpretation of subsurface lithology and reservoir properties across uncored intervals. By combining visual observations from cores with continuous wireline measurements such as gamma ray, resistivity, and sonic logs, geoscientists can refine facies models, identify key stratigraphic boundaries, and improve predictions of reservoir quality and extent.

Altogether, these reports enable a more thorough understanding of the subsurface, supporting more accurate geological models and resource evaluations.

An AI Agent was created using Copilot studio. List of sedimentological and Petrographical analysis reports were uploaded to a SharePoint, which was used as a knowledge in the copilot studio.

The agent was able to respond to the following prompts:

Summarize the Reports • Consolidated Summary of the wells along with Introduction containing Objective and material used are displayed. Lithofacies & Sedimentology content displaying the Classification (Based on Dunham or Embry & Klovan), Facies types and depositional environment description is gathered. Texture, framework grains, matrix, Mineralogy, Diagenetic sequence is extracted. Conclusions & recommendations are also displayed...
Compare Across facies Types • A Wells Summary is displayed, along with comparative Summary of Facies code like SPG, SPP, RPF, ASPF, CgSG, SPWP, SW, WBSP/BBSW and the facies name, texture, porosity, permeability, Common grains, diagnosis information are also extracted.
Extract Images • The reports contain the following embedded images like Photomicrographs, core description logs, facies diagrams, Paragenetic sequences and Thin Section Visuals. Few options appear suggesting the next step to download all the images in a zip.
Download the images in a zip file • A file is downloaded with all the images in a zip folder..

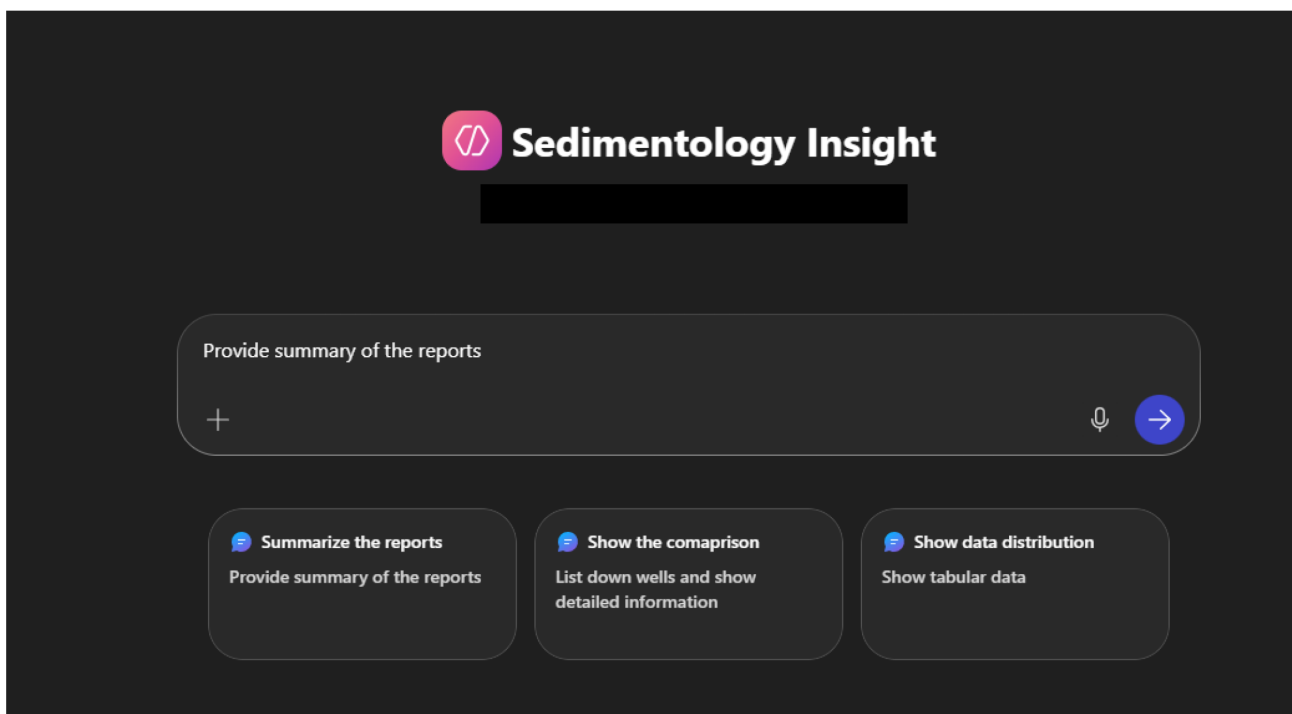


Figure 6—Sedimentology data Insight interface

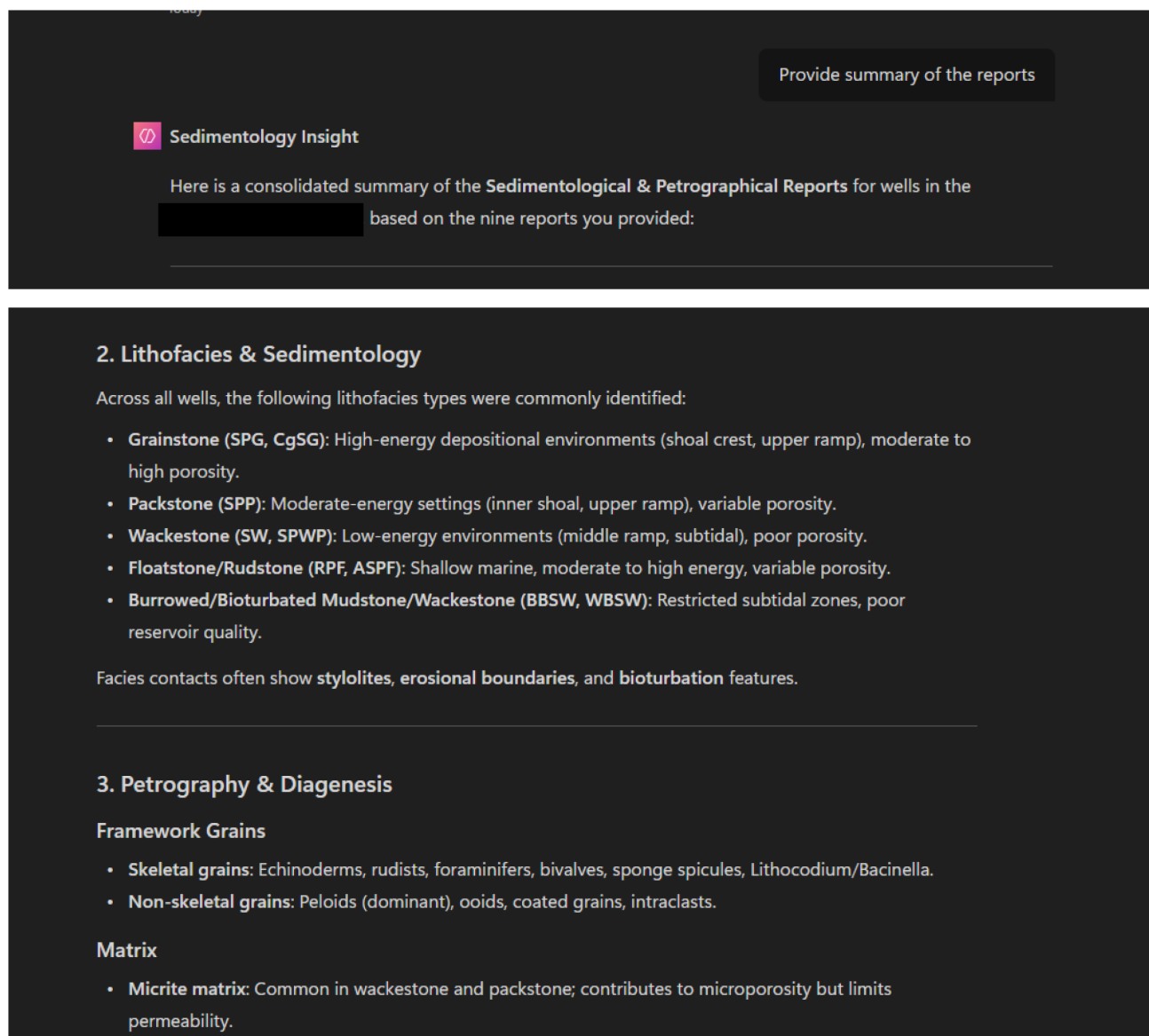


Figure 7—Sedimentology Insight agent shows example of the prompt-response agent for answering queries from reports

Subsurface Insight Agent. The fourth business case showcasing the implementation of an AI Agent center around a comprehensive Portal, designed and refined through ongoing feedback from a diverse range of stakeholders. This Portal consolidates data from multiple sources, including the corporate database, interpretation database, data quality database, subsurface technical application projects, and other associated and often unstructured data repositories. These diverse data sources were meticulously identified, categorized, and integrated, breaking down traditional silos to establish a unified, accessible environment.

A notable feature of the Portal is its knowledge component, which serves as a hub for technical academies' content and a repository of technical training videos. This rich knowledge base can be efficiently shared with users through Portal, empowering teams with relevant, up-to-date information to support decision-making and technical excellence.

To ensure robust data discovery and governance, Portal provides business users with a seamless platform for searching and retrieving relevant datasets. All data is managed under strict governance protocols, guaranteeing both data quality and compliance with organizational standards.

At the heart of the Portal is a Customized AI Agent, intricately integrated to perform a suite of specialized tasks autonomously. This agent is steadily becoming indispensable across a broad spectrum of subsurface

and technical applications. The Subsurface Insight chatbot is a key innovation within this ecosystem, leveraging Portal's SQL database to provide intelligent, context-aware answers to user queries.

The technical foundation of this AI Agent is built on advanced tools such as Lang chain, Ollama, and Chroma DB—a vector database used for efficient storage and retrieval of unstructured data. Lang chain enables the development of autonomous AI agents powered by Large Language Models (LLMs). These agents use reasoning engines to determine the most appropriate actions, which can then be iteratively refined by feeding the outcomes back into the LLM for increasingly accurate results.

Ollama plays a pivotal role by running the web API locally, ensuring that all data interactions remain private and secure within the organization's infrastructure. To facilitate seamless communication between the AI models, the SQL database, and the broader Portal ecosystem, libraries and wrappers—such as the Langchain wrapper for Ollama—are employed. This setup allows users to interact with the SQL database using natural language, making data access intuitive and reducing the technical barrier for non-specialist users.

The database itself is regularly updated to incorporate new datasets and evolve business requirements, ensuring that the Portal remains a dynamic, living resource. The combination of AI-driven agents, robust data integration, and user-centric design culminates in a powerful solution that enhances productivity, promotes knowledge sharing, and supports the strategic objectives of the organization.

The chatbot answers following prompts:

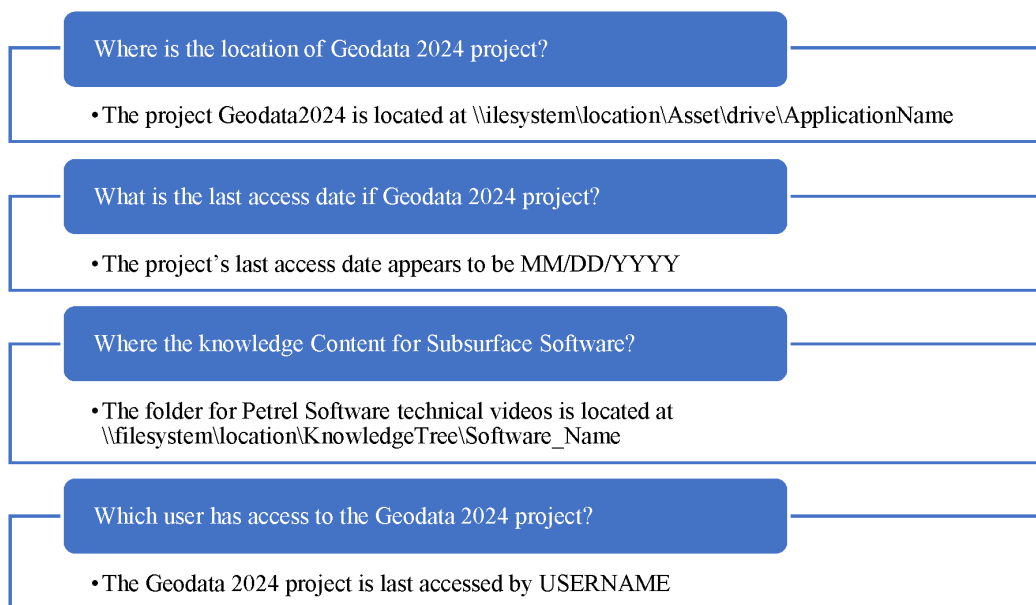


Figure 8—Customized AI Chatbot workflow for specialized agents used as a basis for the Subsurface Insight Agent

The above diagram explains how the chatbot agent works, it handles user queries by intelligently routing them with help of a main controller agent to the specialized agents. When a user submits a question, it is first received by a controller agent, which determines whether the query is related to unstructured or structured data. If the query involves questions related to unstructured data or pdf documents, it is sent to RAG agent which uses Retrieval Augmented Generation method. This agent is using Embedding model to store the document data into small chunks of numeric vectors by diving the main document into multiple chunks and stores them in a Vector Database. A Large Language Model (LLM) then retrieves relevant information from this vector database to generate a natural language response. For structured data, the query is routed to an SQL Agent which translates the question into a standard SQL query which is processed by the SQL Database, retrieves the data from the database tables and then LLM uses this result sets and converts into a user-friendly response natural language response.

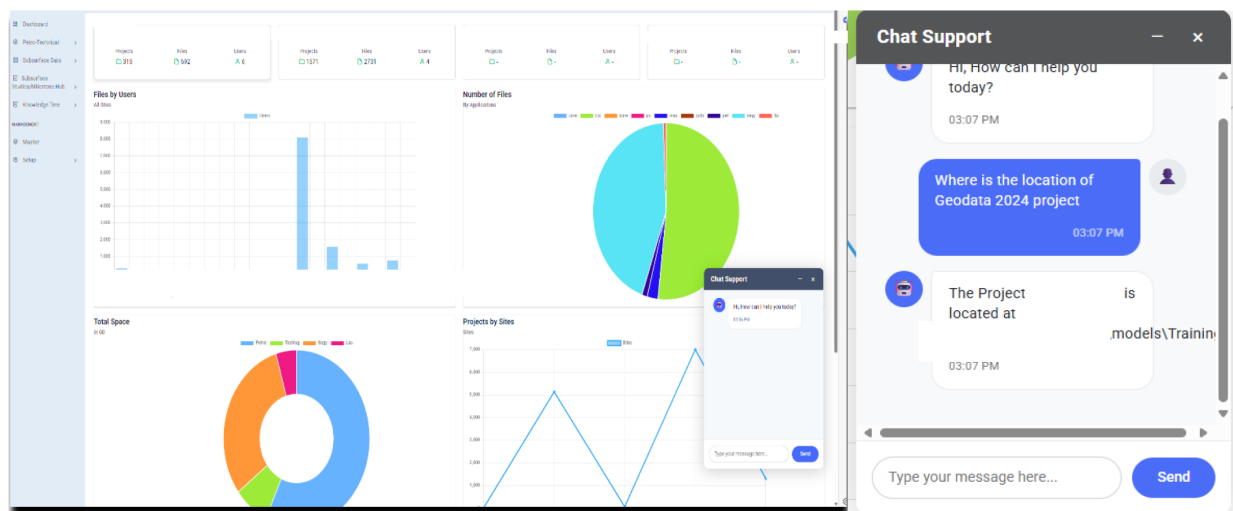


Figure 9—Example of chatbot response created using Ollama framework in custom environment

Subsurface Data Workflows

Analyst Agent. The Analyst agent comes with copilot studio. It executes codes to provide answers, and it shows the thought process. The Analyst Agent, is integrated and represents a significant advance in the landscape of digital productivity and AI-powered assistance. Acting as a bridge between user queries and actionable insights, this agent not only interprets questions but also determines the most appropriate tools and processes needed to deliver comprehensive solutions.

At its core, the Analyst Agent leverages large language models (LLMs), specifically built on OpenAI's o3-mini reasoning mode. This advanced reasoning capability allows the agent to process complex instructions, infer context, and draw on a wide range of data sources to craft relevant, precise, and actionable responses. This involves parsing the language, identifying key terms, and referencing integrated knowledge bases or databases relevant to the domain.

Once the agent identifies the nature of the task, it autonomously selects and invokes the essential tools and resources. For example, if a query requires generating a report from a dataset, the agent will access the appropriate data, execute the necessary code or scripts, and synthesize the findings into a coherent summary. If the task involves providing technical guidance on a particular application, the agent consults repositories of documentation, tutorials, and even past user interactions to furnish a tailored response.

Once a data file is uploaded in the Analyst, it performs coding & executing for data loading, review, analysis, inspection, exploratory analysis and provide finalizes analysis summary. It is further used to plot charts, histograms and predict the data value.

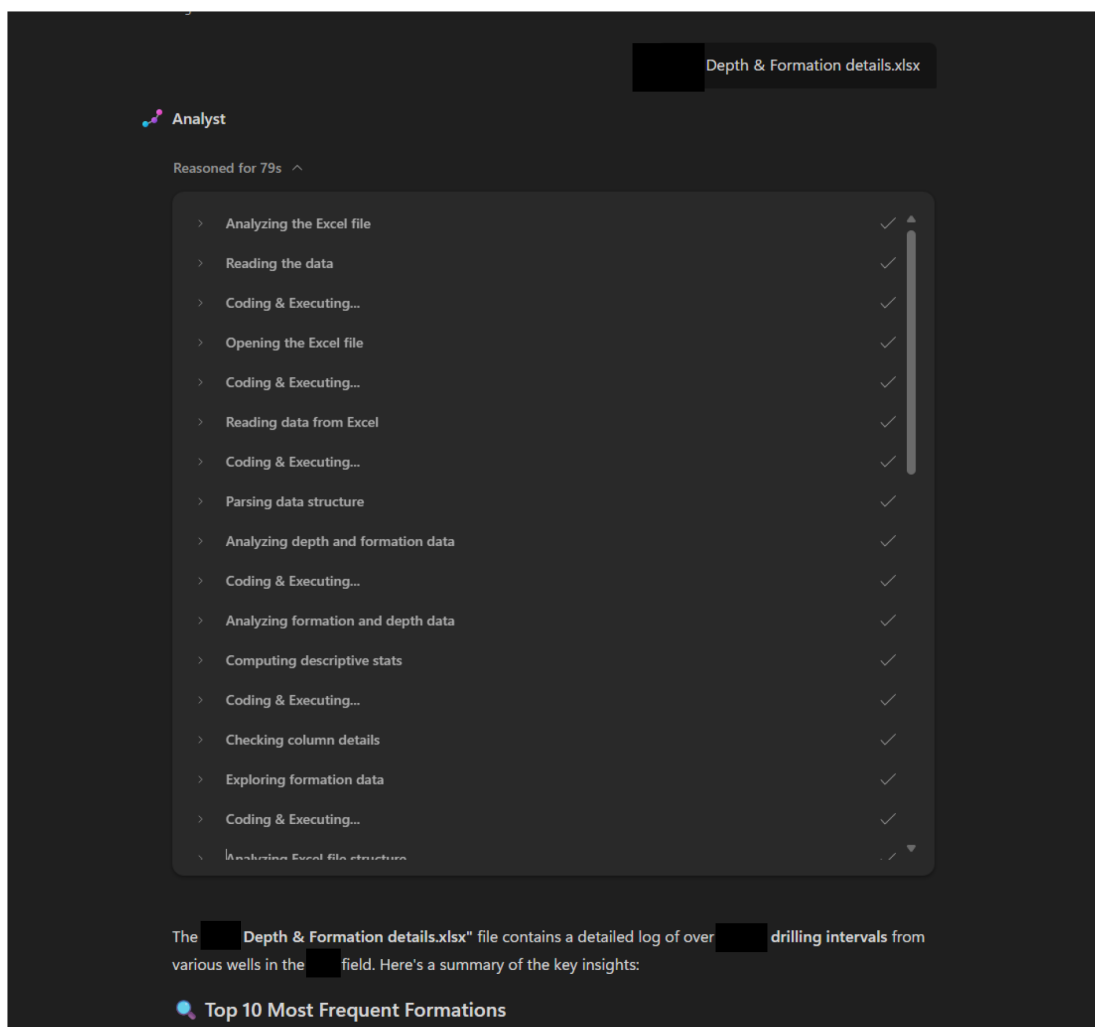


Figure 10—Analyst agent response when data is uploaded and processed

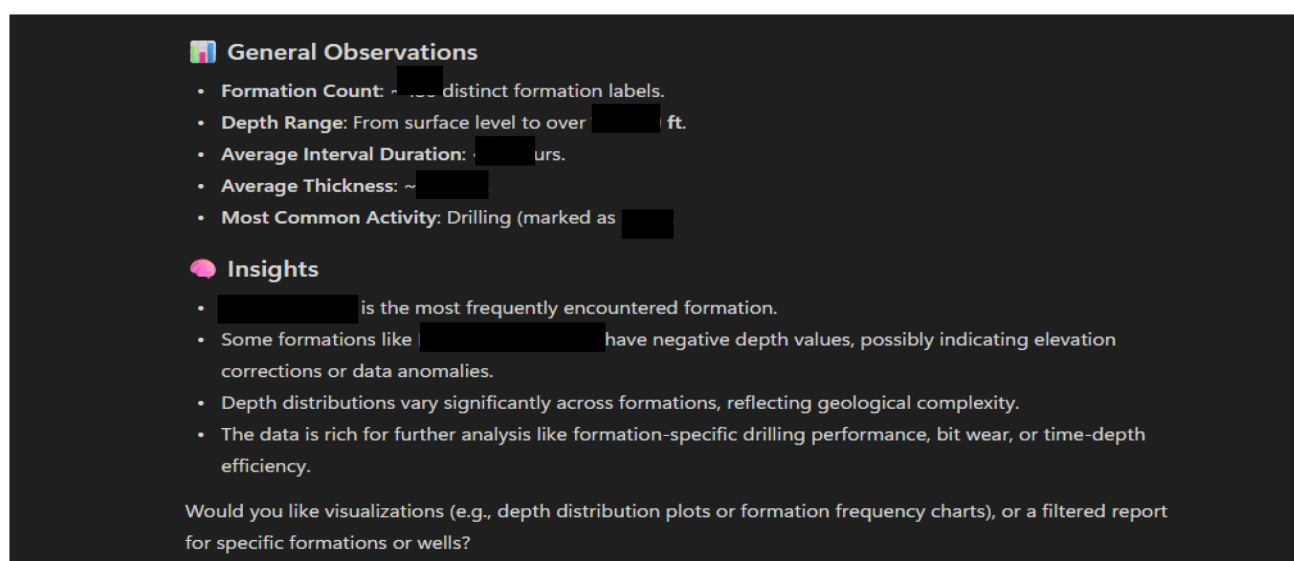
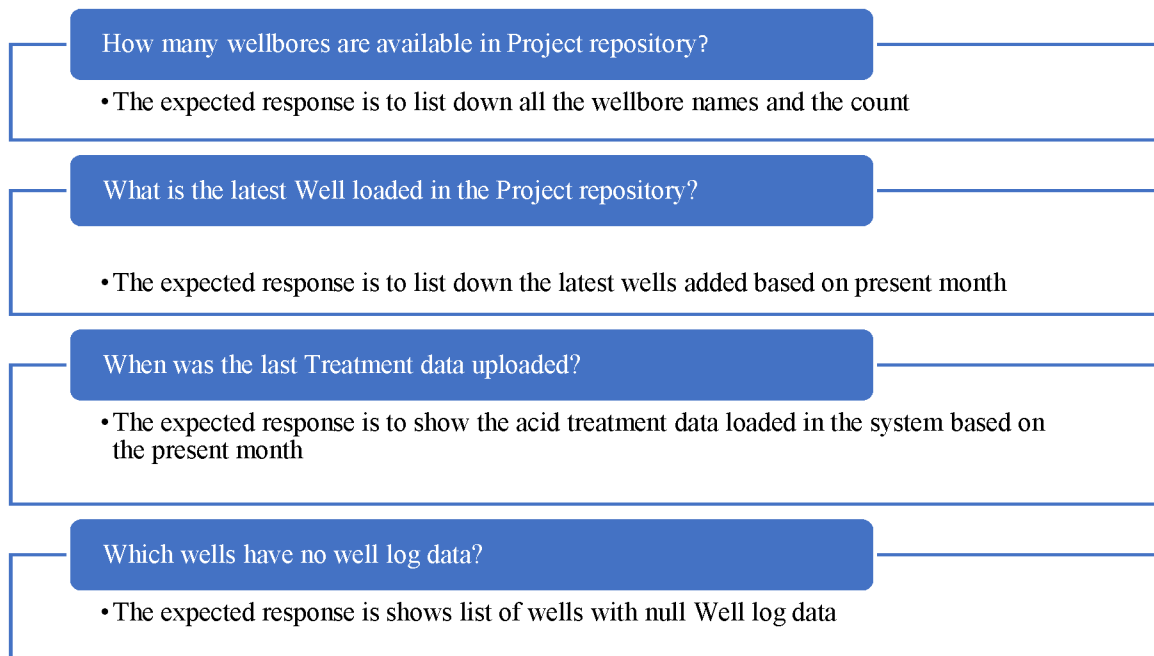


Figure 11—Analyst agent is used to analyze specific data and show insights

Studio Insight Agent. There is another business case which involves creating an AI agent on the SQL database holding data from the Geoscience domain. In the Subsurface domain, Geoscience data types have

the largest volume. With the changes in business requirements, the size of data is growing exponentially. Most Petro-Technical applications used work in siloes. Studio collaboration tool is present to facilitate data sharing and knowledge transfer between the Petrel subsurface software and Techlog wellbore software. It enhances multidisciplinary teamwork by providing a centralized environment for managing and accessing data, promoting efficient workflows across different domains. With the increase in number of wells, the associated data size also grows in the database.



But the accuracy of the prediction is less and the custom model using the Ollama framework cannot show exact information. More advanced and precise configurations are required.

Subsurface Software's Workflow Agent. In the evolving landscape of Geoscience workflows, the integration of AI agents is opening up significant opportunities to streamline, automate, and enhance daily operational efficiency. These workflows are integral to the success of subsurface exploration and development, encompassing a variety of repetitive yet highly specialized tasks that traditionally demand manual input from geoscientists, engineers, and other domain experts.

For instance, well operations constitute a major component of day-to-day activities within the Geoscience domain. Geoscientists routinely load vast volumes of operational data into centralized repositories, interpret new well logs, and update models as drilling progresses. One of the most critical and data-heavy activities is geosteering, which requires real-time analysis and interpretation of wellbore trajectory and geological properties to optimize well placement. The process is highly iterative and involves continuous data assimilation—ranging from mud logs to real-time MWD (Measurement While Drilling) and LWD (Logging While Drilling) data streams. Here, an AI agent, trained in historical data well and equipped with advanced pattern recognition capabilities, could automate initial interpretation steps, flag anomalies, and offer predictive insights for optimal drilling paths, thereby reducing human error and accelerating decision-making.

Similarly, static modeling workflows, such as those used in regional or field development studies, involve frequent updates as new data becomes available. Static geomodels are routinely updated with fresh well logs, seismic interpretations, and production data. Multidisciplinary teams of geologists, petrophysicists, reservoir engineers collaborate to ensure the model accurately reflects the subsurface reality. An AI agent

can greatly assist by automating the ingestion and QC (quality control) of new data, suggesting model updates based on learned geological patterns, even generating preliminary maps or cross-sections for expert review.

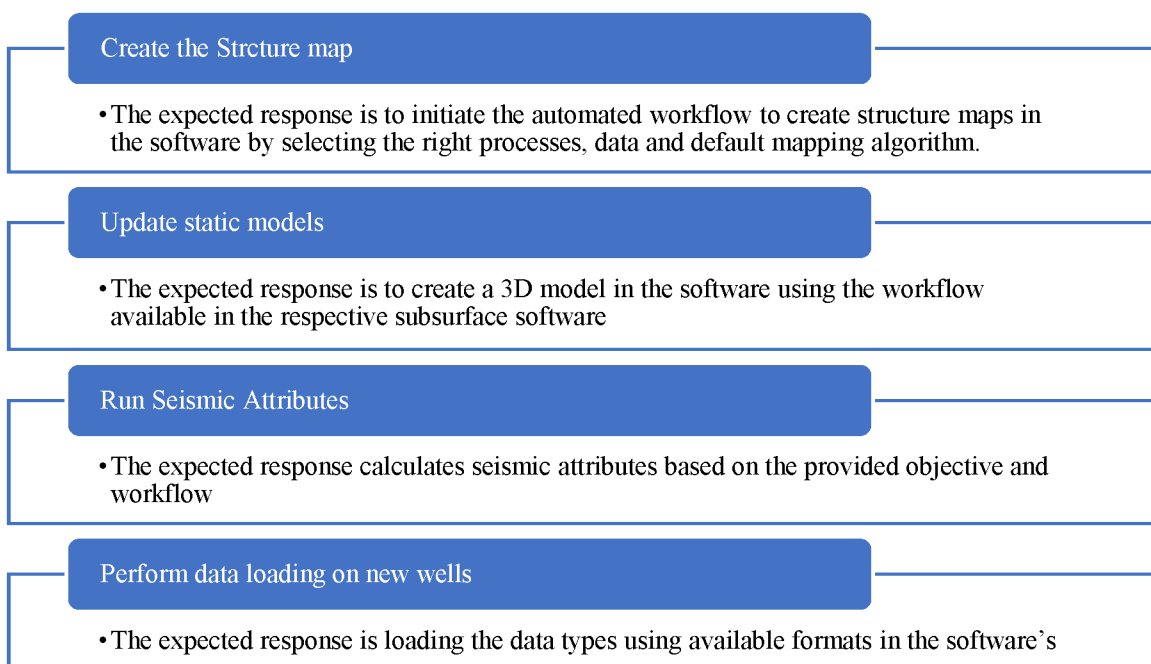
Surface mapping is another area characterized by intensive data processing and cross-domain checks. Each discipline within Geoscience, whether dealing with sedimentology, structural geology, petrophysics, or reservoir engineering adds their layer of interpretation, culminating in comprehensive surface and subsurface maps. The manual compilation and validation of these maps are time-consuming. AI agents, leveraging deep learning and computer vision, could automate the integration of disparate datasets, highlight inconsistencies between layers, and propose harmonized versions, thus enhancing collaboration and reducing turnaround time.

The implications of such AI-driven workflow automation are profound. By offloading routine, repetitive, and data-intensive tasks to intelligent agents, geoscientists are freed to focus on higher-level interpretation and strategic decision-making. AI agents can also ensure greater consistency across projects, facilitate institutional knowledge retention by learning from past workflows, and foster cross-domain innovation through seamless data integration.

Moreover, as the volume and complexity of subsurface data continue to grow, driven by increases in the number of wells, advances in sensor technologies, and the need for real-time operations the scalability and adaptability of AI-powered workflows become even more crucial. A well-configured AI agent not only automates current workflows but also adapts to emerging business requirements, supporting the digital transformation journey of the Geoscience domain.

The promise of AI in automating and enhancing Geoscience workflows lies in its ability to handle large datasets, learn from historical patterns, and augment human expertise, ultimately leading to more efficient, accurate, and innovative subsurface solutions.

The AI Agent is expected to perform the following tasks:



By deploying the AI Agent to handle repetitive and time-consuming tasks, organizations unlock significant value for geoscientists and business users alike. Automation of routine processes means that professionals can redirect their focus toward high-impact activities such as data interpretation, strategic planning, and innovation. This not only increases overall productivity but also elevates the quality of

decision-making across projects. Additionally, the AI Agent's reliability in managing large volumes of data reduces the risk of human error and ensures greater consistency in outputs. Over time, this fosters a more agile and knowledge-driven work environment, empowering teams to adapt quickly to evolving challenges and opportunities within the geoscience sector.

Discussions, Conclusions and Results

The results observed from implementing AI agents that utilize unstructured data is a step forward in Subsurface data analysis and workflow automation. These agents, powered by large language models (LLMs), display proper response and ease in extracting insights, generating summaries. Their ability to process varied formats ranging from text documents to images and well logs ASCII files make available to business users. AI Agentic approach has long term impact on the Subsurface Data and workflows. The flexibility of the AI agent to adapt as new data sources and business requirements arises ensures its continued relevance, aligning with best practices in standardizing data structures. (Pranav Kumar et al 2018).

By deploying the AI Agents to handle manual tasks, organizations can unleash significant value for the Subsurface business users. The automation of data review and extraction such as content generation from the unstructured data allows different disciplines to focus much on decision making processes. This leads to an increase in productivity enhanced quality across daily operations, projects, programs and portfolios.

The collaborative approach of these AI agents supports a single source of truth when it comes to sharing both knowledge and insights. A well-trained agent can be accessed by everyone, breaks down the silos and prevents duplications. This journey simplifies answers to domain specific questions, generating reports, outlier analysis, and predictive analytics based on historical datasets (Chaturvedi et al., 2022 and Schlumberger, 2020).

In a controlled environment and by configuring the private cloud gives flexibility to the users to develop with respect to organizational policies, compliance standards, and data privacy requirements. This compliance ensures that sensitive information is managed with care and that outputs align with both internal guidelines and external regulations, which is an essential feature, as agents become integral to core business processes.

Further, by the integration of vector database, storing and retrieving of the information becomes smoother. The agents can efficiently compare, cluster, and cross-reference relevant data, regardless of its original format or source. This eliminates the idea of uploading the same documents and managing multiple versions.

By following the recommendations and best practices of Agentic AI, organizations can evolve and utilize knowledge sources for better decision making

Acknowledgements

The authors would like to acknowledge the Support of the Management from Data management and Digital Department at Adnoc Onshore.

References

- Rauf Iqbal, Pranav Kumar, Muhammad Aamir, Shaikha Obaid Al-Naqbi, Salahuddin Abdullah Jan and Pravin Kumar, (2022) "IBTIKAR Digital Lab-A Collaborative approach towards research and Development Challenges in Oil and Gas Upstream", SPE-211115-MS, ADIPEC, 31-Oct to 3 Nov 2022.
- Pranav Kumar, "Standardizing the Data Structure and Formats to Meet the Business User requirement". SPE-192834-MS, ADIPEC, November 12-15, 2018.
- Pranav Kumar, Alexey Tveritnev, Salahuddin Abdullah Jan, and Rauf Iqbal "Challenges to Opportunity: Getting Value out of Unstructured data management", SPE-214521-MS, GOTECH, 13-15 March, 2023
- Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., ... & Polosukhin, I. (2017). Attention is all you need. Advances in Neural Information Processing Systems, 30.

- Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics, 4171–4186.
- Chaturvedi, A., Chaturvedi, R., & Prasad, R. (2022). Application of Artificial Intelligence in Oil and Gas Industry. *Springer Series in Geomechanics and Geoengineering*.
- Schlumberger. (2020). Petrel E&P Software Platform Technical Documentation. Schlumberger Information Solutions.
- ADNOC. (2024). Subsurface Data Management Best Practices. ADNOC Technical White Paper Series. <https://python.langchain.com> <https://ollama.com> <https://docs.trychroma.com>
- Society of Petroleum Engineers (SPE). (2023). Best Practices for Subsurface Data Workflows. *SPE Technical Paper Series*. <https://www.microsoft.com/en-us/microsoft-copilotstudio/> <https://aws.amazon.com/what-is/langchain/?nc1=hl> <https://market-news24.com/tag/technology/page/8/>